

Working increment to New geometry §§13–15 and GEOMETRY + GQG Working Master

Geometry of Typed Defects

v0.2 · 11 September 2026 · CC0 / Anonymous

The non-descent locus is one set. Defects are not one kind. This increment adds the locus calculus (product, coarsening, refinement, restriction, predictive identification) and five named defect kinds matching the present corpus. It does not assign a shape to $N_W(\pi)$. The no-shape lemma remains in force.

Unchanged: chemistry freeze, Hidden Quotient v1.7, 39-screen algebra, toroidal all-orders theorem, L=3 PASS, empirical status labels.

0. Standing objects

Declared domain D , total map $\pi:D \rightarrow Q = \pi(D)$, witness $W:D \rightarrow Y$. $N_W(\pi)$ is the set of attained labels whose fibers are heterogeneous for W . Sector space $S_W = \text{im}(\pi, W)$. Canonical refinement $\pi_W(x) = (\pi(x), W(x))$. Fiber diameter δ_W when Y is metric. Any subset of Q can occur as $N_W(\pi)$.

1. Calculus of loci

1.1 Product of witnesses — Lemma D1

For $W = (W_1, W_2)$, $N_{(W_1, W_2)}(\pi) = N_{W_1}(\pi) \cup N_{W_2}(\pi)$. Joint recovery fails on the union. Cleaning one witness does not clean the other.

1.2 Coarsening the witness — Lemma D2

If $W' = f \circ W$ then $N_{W'}(\pi) \subseteq N_W(\pi)$. A coarser question can descend after a finer question has failed. This is New geometry §5 (E_{opt} vs (E_g, E_b)) and §15.1 (Semaev relation vs specified signed output).

1.3 Refining the observation — Lemma D3

If $\pi = r \circ \pi'$ then $r(N_{W'}(\pi')) \subseteq N_W(\pi)$, and no new coarse defect can appear: if $q \notin N_W(\pi)$, no point of $r^{-1}(q)$ lies in $N_{W'}(\pi')$.

1.4 Restricting the domain — Lemma D4

$N_{W|C}(\pi|C) \subseteq N_W(\pi) \cap \pi(C)$. The inclusion can be strict. Descent on C is not descent on D . Record the restriction separately (New geometry §15.3).

1.5 Predictive locus is a static locus — Lemma D5

A deterministic $U:D \rightarrow D$ admits an autonomous retained update on Q iff $N_{\pi \circ U}(\pi) = \emptyset$. Matching a coarser one-step output B is $N_B(\pi) = \emptyset$, which does not imply autonomy of π . Sophie Germain on $n \mapsto 2n+1$ is D5: $r(3) = r(5)$ while $r(U(3)) \neq r(U(5))$. It is not a new closure theorem.

2. Five defect kinds

Each kind has a test, a legal repair, and an illegal repair. A new example must name its kind or declare a sixth.

Kind	Test	Legal repair	Illegal repair
I Operation loss	W_{rel} descends, W_{out} does not, same D and π	Retain the coordinate that splits the out-fiber (sign / y)	Treat vanishing of W_{rel} as recovery of W_{out}
II Unit / character loss	π kills a unit that W (a log or character) sees	Add a character with declared rank, kernel, modulus, precision	Promote one separating coefficient to a global theorem; omit $\gcd(h, \ell) = 1$

III Subgroup injectivity	$\ker \varphi \cap C = \{e\}$ while $\ker \varphi \neq \{e\}$	Restrict the source to C and record D4	Infer injectivity on G, a speedup, or GHS/HCDLP
IV Discrete sector	$A_W = 0$ on components, component constants disagree	Retain a sheet label, or pass to S_W	Treat vanishing vertical differential as descent on disconnected fibers
V Representability gap	$N_W(\pi) = \emptyset$ but a needed object lies outside $\text{im } \pi$	Name the missing object; enlarge observation or codomain	Call the gap $N_W(\pi)$. Collision $\neq$ missing dual

Kind I instance: $E(F_{13})$, $P=(2,1)$, $Q=(3,3)$, $-Q=(3,10)$; π keeps x-coordinates; $x(P+Q)=12 \neq x(P-Q)=11$. Kind II instance: 1 and $1+\sqrt{2}$ generate the same ideal in $\mathbb{Z}[\sqrt{2}]$; reduction $\sqrt{2} \mapsto 3$ into F_7 separates base-3 logs mod 3. Pair separation is not Leopoldt.

3. Identification table

A	B	Identify?
$N_W(\pi) = \emptyset$	existence of reduced witness W	Yes — Thm 14.2
$N_{\{\pi \circ U\}}(\pi) = \emptyset$	autonomous retained update	Yes — D5
$N_{\{f \circ W\}}(\pi) = \emptyset$	$N_W(\pi) = \emptyset$	No — D2 is only $\subseteq$
N on a restriction C	descent on D	No — D4
$q \in N_W(\pi)$	$q \in \partial N_{\{W, \varepsilon\}}(\pi)$	No — §14.8
source-space transition	quotient-space boundary	No — lithium pressure
Kind I existential zero	Kind I specified output	No
Kind II pair character	global Schirokauer sufficiency	No
Kind V missing dual	fiber collision	No
combining projects	transferring proof	No

4. Sector space and the predictive stack

Kinds I–IV are multiple points in an r -fiber of $S_W \rightarrow Q$. The kind is the extra coordinate that splits that fiber. Kind V is a graph of a function plus a missing object outside the image. Canonical refinement π_W kills I–IV for this W by retaining W ; it is not a mechanistic coordinate and not a runtime predictor.

Declare in order: static W and N_W ; update or kernel; predictive witness $\pi \circ U$ or the next-observation law (D5); horizon- h diameter δ_h^B ; clock only if fixed-time descent holds and cross-time fails. Do not feed future B-values into the present predictor. 39-screen and TD-COS-FH-001 instantiate D5 on different (D, U, W) . They are not special cases of each other.

5. Disposition

TYPED DEFECT CALCULUS: ESTABLISHED
GLOBAL NON-DESCENT SHAPE: STILL NOT MEASURED
SOURCE FREEZES: UNCHANGED

Point Working Master §1 here. Keep proofs in the companion markdown. Reuse the arithmetic packet only with New geometry §15.5 qualifications: probable primality, Semaev resultant factors, prime-degree Weil restriction still has a base field, unit-character rank/kernel/precision, and h^{-1} requires $\gcd(h, \ell) = 1$.